

Assignment 3

RAG FYP Handbook Assistant

Submitted by:

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1. System Overview

The system is a text-only RAG (Retrieval-Augmented Generation) pipeline designed to answer questions based on the **BS Final Year Project Handbook 2023**. It consists of an ingestion script, a retrieval engine, and a Streamlit-based user interface.

2. Technical Settings

Setting	Value
Chunking Strategy	RecursiveCharacterTextSplitter
Chunk Size	1000 characters
Chunk Overlap	100 characters
Embedding Model	nomic-embed-text (via Ollama)
Vector Database	ChromaDB (Local persistence)
LLM	Phi-3-mini (via Ollama)
Retrieval (k)	5
Similarity Threshold	0.2 (Cosine similarity relevance score)

3. Implementation Details

Ingestion (ingest.py): Uses `pdfplumber` for robust text extraction from the PDF. It preserves page numbers in metadata for grounding.

Retrieval (ask.py): Computes cosine similarity between query and document embeddings. Chunks with relevance scores below the threshold are discarded.

Grounding: The prompt instructs the LLM to answer *ONLY* from the provided context and cite specific page numbers using the format (p. X).

4. Validation Q&A Examples

Example 1: Margins and Spacing

Question: What margins and spacing do we use?

Answer: Top margin = 1.5", bottom margin = 1.0", left margin = 2.0", right margin = 1.0", line spacing = 1.5, paragraph spacing = 6 pts (p. 40) as per the BS Final Year Project Handbook 2023.

Example 2: Endnotes Usage

Question: How should endnotes like "Ibid." and "op. cit." be used?

Answer: To use "Ibid." (citing the same source), if only page numbers differ, add them after Ibid., e.g., "Ibid. p. XX". For multiple sources with intervening references, use "op. cit." followed by the page number (p. 41).

5. UI Screenshot (Mockup Description)

The Streamlit interface (app . py) features:

- A clean header with a book icon.
- A prominent text input for student queries.
- An "Ask" button with a processing spinner.
- An answer panel that highlights the generated response.
- A collapsible "**Sources (page refs)**" section showing the exact text chunks used for the answer along with their similarity scores.

Example 3: Thesis Chapters

Question: How many chapters should be in the thesis?

Answer: The thesis should be organized into five chapters: 1. Introduction, 2. Background and Related Work, 3. Methodology, 4. Implementation and Results, and 5. Conclusion and Future Work (p. 50).

Example 4: Proposal Deadline

Question: When is the proposal deadline?

Answer: The proposal deadline is September 15, 2023 (p. 46).

Example 5: Plagiarism Severity

Question: What happens if I commit plagiarism?

Answer: Plagiarism is a serious academic offense and may result in failure of the course or suspension from the university (p. 27).